Algorithmic Pseudocode Sample 3

Source-backed Image2Struct algorithm sample

Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

for $t \in \{-1, ..., -T^{\text{traceback}}\}$ **do** ▷ *Retrieve the values of* $P_{u,t}$
  $P_{u,t} \leftarrow \text{Power.GetValue}(t)$
end for
for $t \in \{-1, ..., -T^{\text{traceback}}\}$ **do** ▷ *Initial conditions on the state variables*
  if $P_{u,t} > \text{MinimumPower.GetValue}(t)$ **then**
    $S_{u,t}^{\text{OFF}} \leftarrow 0$
    $S_{u,t}^{\text{STOP}} \leftarrow 0$
    $S_{u,t}^{\text{START}} \leftarrow 0$
    if $P_{u,t} < P_{u,t-1}$ **then** ▷ *Exact initialization required only if the FLAT state is defined.*
      $S_{t-1}^{\text{ON_UP}} \leftarrow 0$
      $S_{t-1}^{\text{ON_DOWN}} \leftarrow 1$
      $S_{t-1}^{\text{ON_FLAT}} \leftarrow 0$
    else if $P_{u,t} > P_{u,t-1}$ **then**
      $S_{t-1}^{\text{ON_UP}} \leftarrow 1$
      $S_{t-1}^{\text{ON_DOWN}} \leftarrow 0$
      $S_{t-1}^{\text{ON_FLAT}} \leftarrow 0$
    else if $P_{u,t} = P_{u,t-1}$ **then**
      $S_{t-1}^{\text{ON_UP}} \leftarrow 0$
      $S_{t-1}^{\text{ON_DOWN}} \leftarrow 0$
      $S_{t-1}^{\text{ON_FLAT}} \leftarrow 1$
    end if
  else if $P_{u,t} > 0$ **then** ▷ *Reconstruct the startups and shutdowns*
    if $P_{u,t} < P_{u,t-1}$ **then**
      $S_{t}^{\text{STOP}} \leftarrow 1$
      $S_{t}^{\text{START}} \leftarrow 0$
    else
      $S_{t}^{\text{STOP}} \leftarrow 0$
      $S_{t}^{\text{START}} \leftarrow 1$
    end if
  else ▷ *Final possibility: the unit is OFF.*
    $S_{u,t}^{\text{OFF}} \leftarrow 1$
    $S_{u,t}^{\text{STOP}} \leftarrow 0$
    $S_{u,t}^{\text{START}} \leftarrow 0$
    $S_{t-1}^{\text{ON_UP}} \leftarrow 0$
    $S_{t-1}^{\text{ON_DOWN}} \leftarrow 0$
    $S_{t-1}^{\text{ON_FLAT}} \leftarrow 0$
  end if
end for

Algorithm 2 Initialization of the control and state variables